

PHYSICS — Exam 03 — Unit 5: Pressure and Deformation in Solids

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PERFECT SCORE — 65/65 (100%)

Third consecutive perfect Physics exam. Third perfect score overall (joined Math-01 and Bio-02 re-sit).

TOTAL

65/65

100%

GRADE

A+

100%

SECTION A (MCQS)

12/12

100% — 11th consecutive perfect MCQ

SECTION B (SHORT)

33/33

100%

SECTION C (LONG)

20/20

100%

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Topic	Correct	Seemab	Result
1	Definition of elasticity	B	B	1/1
2	Hooke's law holds up to	C	C	1/1
3	SI unit of spring constant	A	A	1/1
4	SI unit of pressure	D	D	1/1
5	Spring constant (2 kg, 5 cm)	A	A	1/1
6	Most elastic material	D	D	1/1
7	Atmospheric pressure at sea level	B	B	1/1
8	Liquid pressure formula	C	C	1/1
9	Instrument measuring atmospheric pressure	B	B	1/1
10	Pascal's Principle — pressure transmitted	C	C	1/1
11	Atmospheric pressure smallest at	D	D	1/1
12	Hydraulic lift — output force when $A_2=100\times A_1$	A	A	1/1

Section A: 12/12 — All correct. This is Seemab's 11th consecutive perfect MCQ score (132 correct MCQs in a row across 11 exams).

Notable MCQ-6 (most elastic material = Steel, not Rubber): Seemab correctly identified that Steel has the highest Young's modulus. This has tripped up many students who confuse "stretches most" with "most elastic".

Notable MCQ-11 (pressure smallest at altitude = Murree): Correctly identified that Murree at ~2300 m above sea level has the lowest atmospheric pressure among the options.

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)

Q.2(i) MAIN — Elasticity, Elastic Limit, examples

3/3

Elasticity defined precisely: "ability of the deforming body to return to its original shape and size when the deforming force is removed." Full mark.

Elastic limit clearly explained: material will not return to original dimensions beyond this point — permanent deformation occurs. Full mark.

Examples: spring (elastic), clay and dough (inelastic). Both types given. Full mark.

Q.2(ii) MAIN — Hooke's Law: statement, mathematical form, spring constant

3/3

Statement: within elastic limit, extension is directly proportional to restoring force. Correct.

Mathematical form: $F(\text{res}) = -kx$ stated correctly with the restoring-force sign convention.

Spring constant: defined as ratio of restoring force to extension; represents stiffness; SI unit = Nm^{-1} correctly stated.

Q.2(iii) MAIN — Spring constant calculation ($m=60 \text{ kg}$, $x=8 \text{ cm}$, $g=10 \text{ m/s}^2$)

3/3

$F = mg = 60 \times 10 = 600 \text{ N}$ — correct. Unit conversion $8 \text{ cm} \rightarrow 0.08 \text{ m}$ explicitly shown.

$k = F/x = 600/0.08$ — correct formula applied.

$k = 7500 \text{ N/m}$ — exactly matches expected answer.

Q.2(iv) OR — Pressure calculation ($m=50 \text{ kg}$, $A=0.02 \text{ m}^2$, $g=9.8 \text{ m/s}^2$)

3/3

$F = W = mg = 50 \times 9.8 = 490 \text{ N}$ — correct.

$P = F/A$ formula correctly written.

$P = 490/0.02 = 24,500 \text{ Pa} = 24.5 \text{ kPa}$ — correct with unit conversion shown.

Q.2(v) OR — Two applications of atmospheric pressure (straw and syringe)

3/3

Straw: sucking lowers air pressure in mouth below atmospheric; atmospheric pressure on liquid surface is now greater; pressure difference pushes liquid up straw. Mechanism correctly stated.

Syringe: pulling piston back decreases pressure inside cylinder; atmospheric pressure acting on liquid surface outside drives liquid through needle into syringe. Mechanism correctly stated.

Q.2(vi) MAIN — Mercury barometer: construction, working, height represents

3/3

Construction: "long glass tube (almost 80 cm), closed at one end, filled with mercury, inverted into a container (trough) of mercury. The closed end creates a vacuum above the mercury column." Accurate.

Working: "mercury falls until the weight of the mercury column is exactly balanced by atmospheric pressure pushing down on mercury in the trough. Works on principle of hydrostatic equilibrium." Accurate.

Height represents: $h = 760 \text{ mm Hg} = 76 \text{ cm} = 1 \text{ atm} = 101.325 \text{ kPa}$ — all values correctly stated.

Q.2(vii) OR — Submarine pressure calculation ($h=500 \text{ m}$, $\rho=1025 \text{ kg/m}^3$, $g=9.8 \text{ m/s}^2$)

3/3

$P = \rho gh$ formula applied. $P = 1025 \times 9.8 \times 500 = 5,022,500 \text{ Pa}$. Expressed as $5.02 \times 10^6 \text{ Pa} = 5.02 \text{ MPa}$. Fully correct including scientific notation and unit conversion.

Q.2(viii) OR — Four applications of a manometer

3/3

Four distinct applications stated: (1) measuring vacuum, (2) calibrations, (3) leak testing, (4) flow of fluid. All four are valid and map directly to the answer key list.

Q.2(ix) MAIN — Manometer vs Barometer: two differences**3/3**

Three differences given in clean tabular form (more than required). Purpose distinction (fluid pressure relative to atmosphere vs atmospheric pressure of air) is the key 1-mark point. Design (multiple forms vs one basic design) and fluid used (any heavy liquid vs mercury only) both correctly stated.

Three differences provided where two are required — good exam technique to show depth of understanding.

Q.2(x) OR — Hydraulic car brake system: step by step**3/3**

All four components named and explained: (A) Brake pedal exerts force on master cylinder piston; (B) Master cylinder — force increases pressure in brake fluid; (C) Brake lines — increased pressure transmitted equally to all wheels; (D) Brake calipers — pressure acts on calipers/wheel cylinders, applying force to brake pads. Matches answer key exactly.

Q.2(xi) OR — Divers wear special suits + pressure at 8.5 km depth**3/3**

Explanation: at great depth $P = \rho gh$ is enormous; high inward pressure on all body surfaces; without protection, pressure would crush the body, rupture blood vessels, and collapse the lungs. All three consequences named — 2 marks.

Calculation: $P = \rho gh = 1000 \times 9.8 \times 8500$ (8.5 km $\rightarrow$ 8500 m conversion shown) = 8.33×10^7 Pa = 83.3 MPa. Fully correct — 1 mark.

Section B: 33/33 — First ever perfect Section B score in Physics.

SECTION C — Long Answer Questions (4 × 5 = 20 Marks)**Q.3(i) MAIN** — Elasticity, Elastic Limit, Hooke's Law, Force-Extension Graph (4 points)**5/5**

Elasticity (1 mk): defined as "ability of a deformed body to return to its original shape or size when the deforming forces are removed" — textbook-precise wording. Full mark.

Elastic Limit (1 mk): "most materials are elastic up to a certain limit known as Elastic Limit; beyond this, materials will not return to original dimensions when deforming force is removed." Full mark.

Hooke's Law (1 mk): statement, mathematical form $F(\text{res}) = -kx$, and spring constant relationship — all present. Full mark.

Force-Extension Graph (2 mk): correctly drawn with all four required points labeled — A (Proportional limit), B (Elastic limit), C (Ultimate stress), D (Breaking point). Axes drawn. Full 2 marks.

Q.3(ii) OR — Atmospheric pressure: definition, mercury barometer, altitude variation, weather indication**5/5**

Definition (1 mk): "the pressure that atmospheric particles exert on the surface of the earth and all over the surface of objects on it." Sea level value 1.013×10^5 Pa stated. Full mark.

Mercury Barometer construction and working (2 mk): glass tube ~80 cm, closed at one end, filled with mercury, inverted into mercury trough; vacuum at closed end; mercury falls until weight balanced by atmospheric pressure; hydrostatic equilibrium principle stated. Height at sea level 760 mm Hg = 76 cm = 1 atm = 101.325 kPa. Full 2 marks.

Variation with altitude (1 mk): as altitude increases, weight of air column above decreases so atmospheric pressure decreases. Full mark.

Weather indication (1 mk): falling barometer = approaching storms/rain; rising barometer = fair weather; isobars on weather maps explained. Full mark.

Derivation Step 1 (1 mk): cylindrical liquid column with height h , base area A , density ρ ; $V = Ah$ stated; $\rho = m/V \rightarrow m = \rho Ah$. Full mark.

Derivation Step 2 (1 mk): $F = W = mg = \rho Ahg$; $P = F/A = \rho Ahg/A$. Full mark.

Derivation Step 3 (1 mk): $P = \rho gh$ — result clearly stated; "liquid pressure depends only on density ρ and depth h , not on shape or total volume." Full mark.

Drill-hole activity: water shoots fastest and furthest from the lowest hole because pressure is greatest at greater depth. Correct explanation.

Submarine calculation (2 mk): $h = 8.5$ km $\rightarrow$ 8500 m conversion shown; $P = 1000 \times 9.8 \times 8500 = 8.33 \times 10^7$ Pa = 83.3 MPa. Full 2 marks.

Note: the answer key uses $\rho = 1000$ kg/m³ for this calculation (the same value Seemab used). Perfect alignment.

Pascal's Principle (1 mk): "An external pressure applied to an enclosed fluid is transmitted unchanged to every point within the fluid." Verbatim precision.

Hydraulic lift derivation (2 mk): Piston 1 = A_1 (small input), Piston 2 = A_2 (large output); $P_1 = F_1/A_1$; $P_2 = F_2/A_2$; by Pascal's Principle $P_1 = P_2$; therefore $F_1/A_1 = F_2/A_2$; rearranges to $F_2 = (A_2/A_1) \times F_1$. Full algebraic derivation shown.

Calculation (2 mk): Given $A_1 = 0.002$ m², $A_2 = 0.9$ m², $m = 1800$ kg, $g = 9.8$ m/s²; $F_2 = W = 1800 \times 9.8 = 17,640$ N; $F_1 = (A_1/A_2) \times F_2 = (0.002/0.9) \times 17,640 = 39.2$ N. Correct answer.

Good self-correction visible on paper: an initial line shows $F_2 = (0.9/0.002) \times 17640$ (wrong direction), which was crossed out and immediately corrected to $F_1 = (0.002/0.9) \times 17640 = 39.2$ N. The final answer is right. This kind of mid-problem self-correction is excellent exam technique.

Section C: 20/20 — First ever perfect Section C score in Physics.

Comparison — Physics Exams

Exam	Date	Chapters	Section A	Section B	Section C	Total	%
exam-01-ch1-2	2026-04-07	Units 1-2	12/12	32.5/33	19/20	63.5/65	97.7%
exam-02-ch3-4-9	2026-04-10	Units 3,4,9	12/12	32/33	18/20	62/65	95.4%
exam-03-ch5	2026-05-18	Unit 5	12/12	33/33	20/20	65/65	100%

Physics average across 3 exams: **98.3%** (190.5/195). Seemab has now scored 12/12 MCQs across all three Physics exams — 36 consecutive correct MCQs in Physics alone.

Strengths and Areas to Watch

Perfect conceptual precision under exam conditions

Every definition in this paper — elasticity, elastic limit, Hooke's Law, pressure, atmospheric pressure, Pascal's Principle — used textbook-grade wording. No vague approximations.

Calculation discipline: unit conversions every time

cm to m (Q.2(iii)), Pa to kPa (Q.2(iv)), km to m (Q.2(xi), Q.3(iii)). All conversions shown explicitly in working. No shortcuts taken under time pressure.

Mechanism stated, not just outcome

Straw (Q.2(v)): "sucking lowers pressure below atmospheric; atmospheric pressure on surface pushes liquid up."
Syringe: "pulling piston back decreases pressure; atmospheric pressure drives liquid through needle." Both answers describe why, not just what.

Self-correction mid-calculation (Q.3(iv))

Incorrectly set up $F_2 = (A_2/A_1) \times F_1$ at first, caught it, crossed it out, and computed $F_1 = (A_1/A_2) \times F_2 = 39.2$ N. This demonstrates active error-checking, not mechanical formula-plugging.

All four force-extension graph points labeled (Q.3(i))

A (proportional limit), B (elastic limit), C (ultimate stress), D (breaking point) — all four correctly drawn and named. Students commonly omit C or confuse B and C.

Area for ongoing monitoring: Section C completeness across all subjects

This is the first perfect Section C in Physics. Chemistry exam-03 Section C was 13.5/20 (weakest across all attempts). Keep the habit of checking that each long question part is fully addressed before moving on.